

DNS Log Analysis

Splunk Basics – DNS Log Analysis Lab Report

STEP-BY-STEP GUIDE REPORT

OBJECTIVE

The objective of this lab is to understand how **DNS logs can be analyzed using Splunk** to identify network activity patterns.

During this lab we will:

- Ingest **Zeek DNS logs (JSON format)** into Splunk.
- Learn how Splunk indexes and parses log data.
- Use **SPL (Search Processing Language)** to analyze DNS activity.
- Identify:
 - Most queried domain names
 - Most active source IPs
 - Types of DNS queries in the network

This helps security analysts detect **suspicious DNS activity, malware communication, or abnormal traffic patterns**.

TOOLS AND ENVIRONMENT

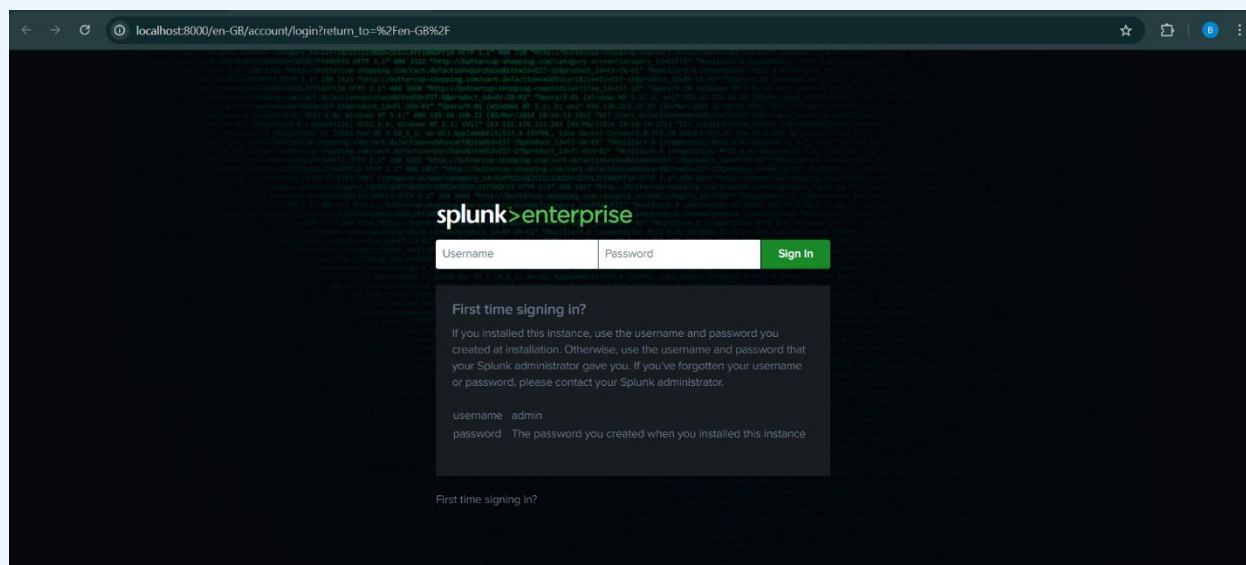
Component	Description
Splunk	Log management and analysis platform
Data Source	Zeek DNS logs
Log Format	JSON
Index Used	dns_lab
Source type	_json

LAB SETUP

Before starting the analysis, the DNS log file must be uploaded to Splunk.

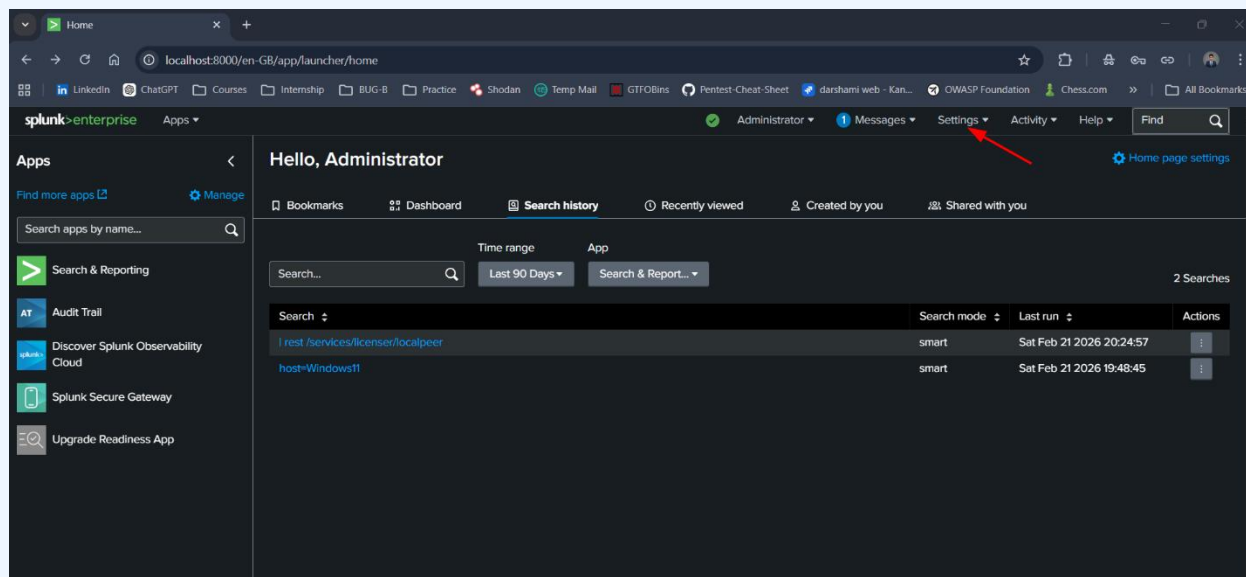
Step 1 – Open Splunk Web

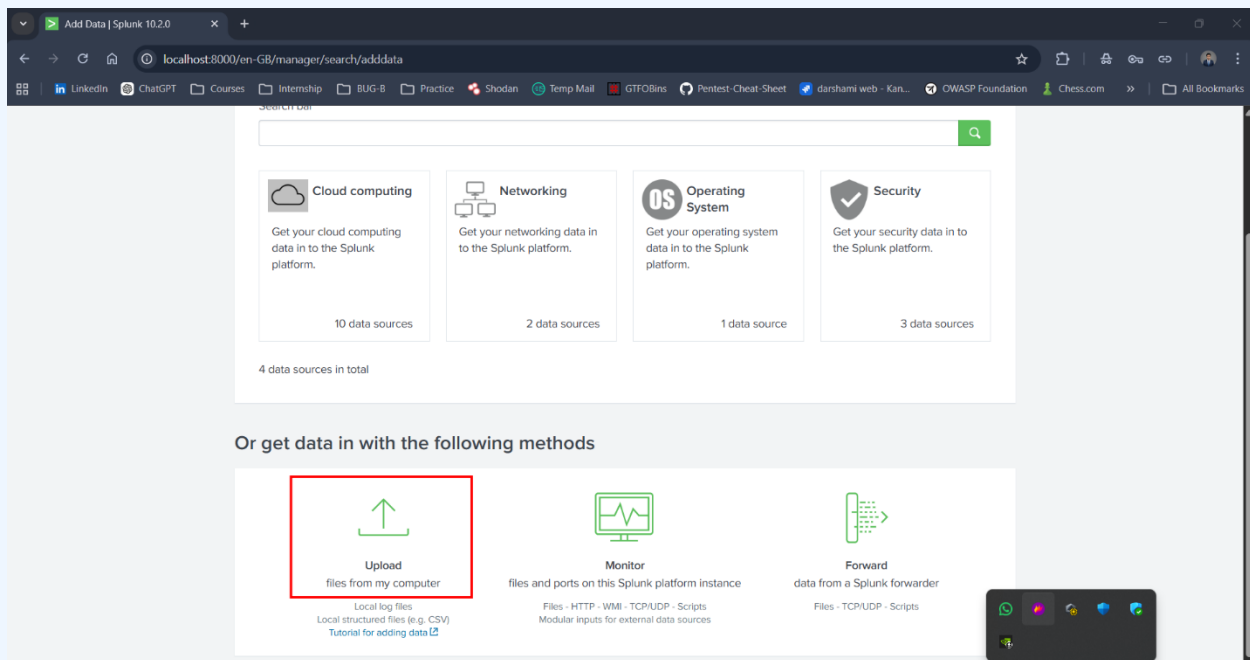
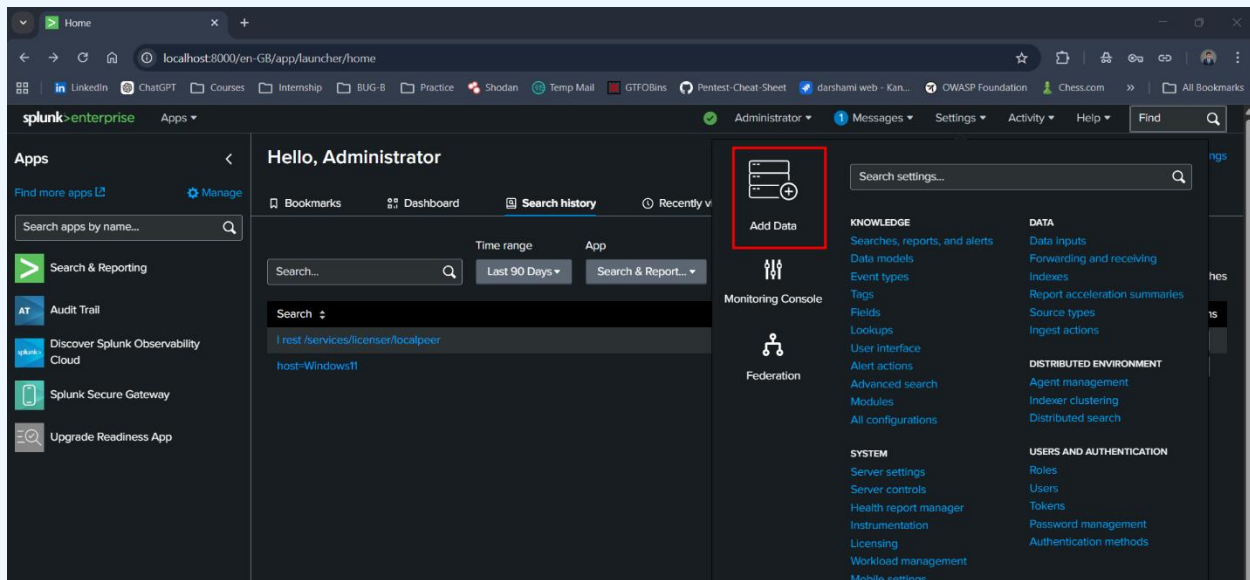
1. Open the **Splunk Web Interface** in your browser.
2. Log in using your credentials.



Step 2 – Add Data

1. Navigate to:
Settings → Add Data
2. Select **Upload**.



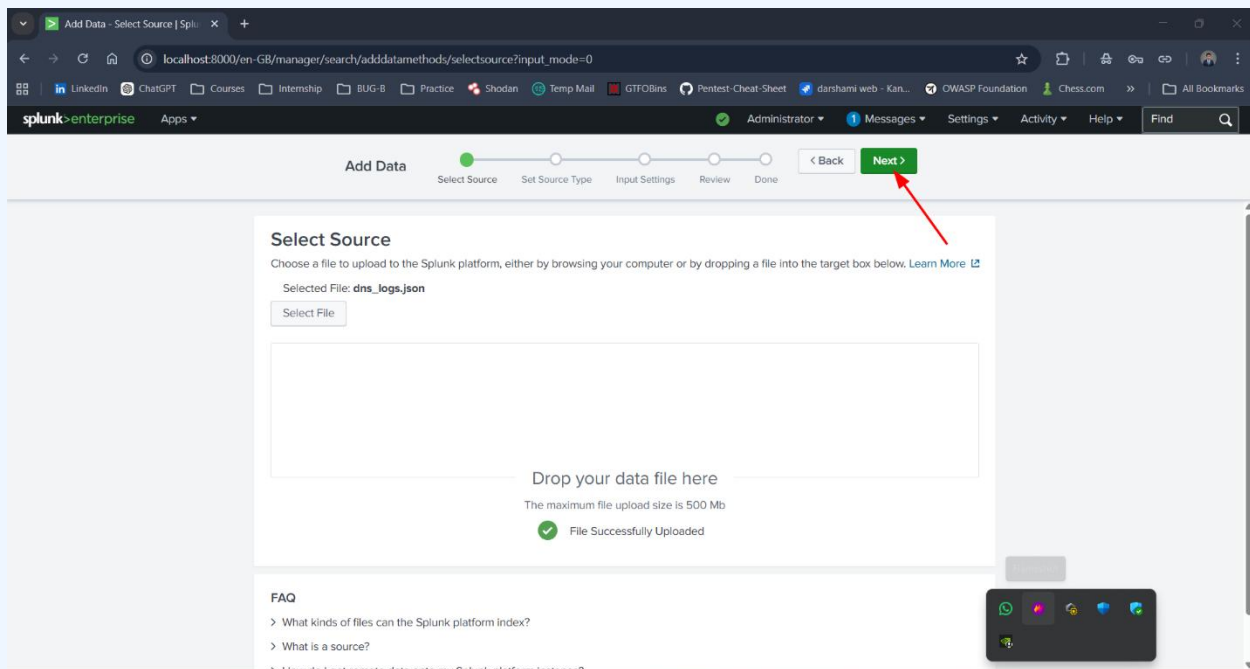
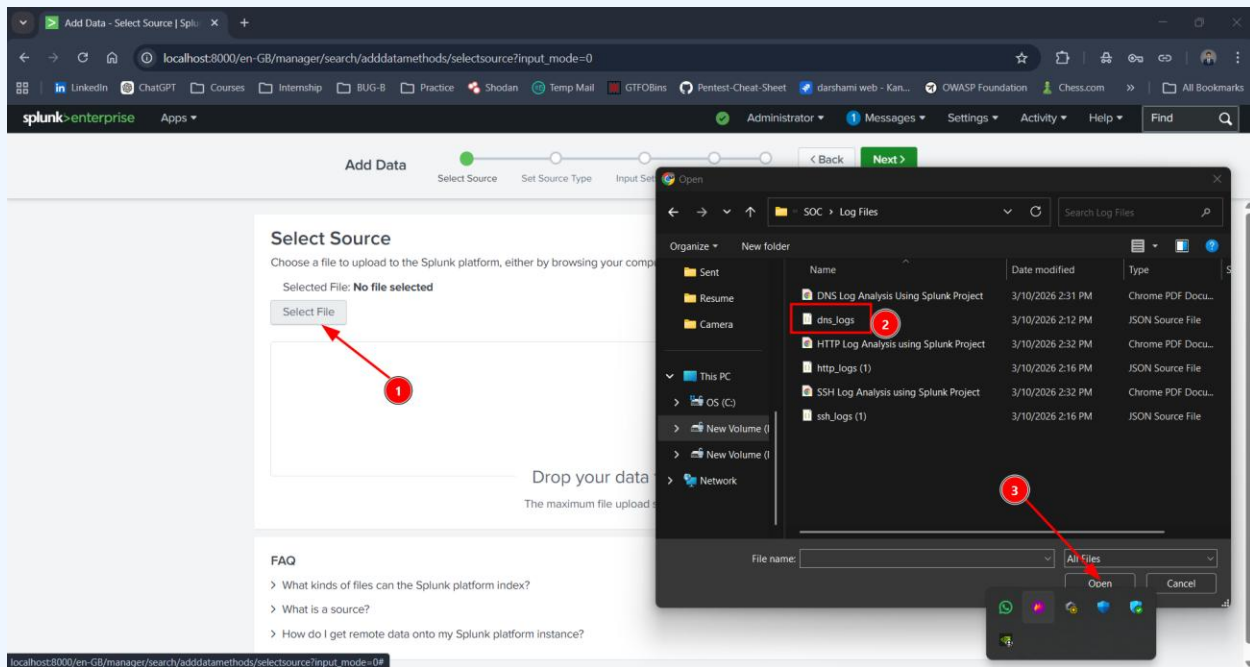


Step 3 – Upload DNS Log File

1. Click **Select File**.
2. Upload the file:

`dns.log`

This file contains **DNS** query records generated by Zeek network monitoring tool.



Step 4 – Configure Source Type

Set the **Source Type** as:

`_json`

Reason:

- Zeek logs are structured in **JSON format**
- Splunk automatically extracts fields like:
 - query
 - id.orig_h
 - qtype
 - answers

The screenshot shows the Splunk 'Set Source Type' configuration page. The 'Source type' dropdown is set to '_json'. A table of sample data is displayed below, showing fields like _time, aa, answers, id.orig_h, id.orig_p, id.resp_h, id.resp_p, proto, qclass, qtype, and query. The 'Next' button in the top navigation bar is highlighted with a red arrow.

	_time	aa	answers	id.orig_h	id.orig_p	id.resp_h	id.resp_p	proto	qclass	qtype	query
1	23/04/2025 14:51:09.069	false	alias.google.com	192.168.112	58167	192.168.11	53	udp	IN	CNAME	google.com
2	23/04/2025 14:51:09.069	false	192.168.101.14	192.168.112	13629	192.168.13	53	udp	IN	A	fileserv.localhost
3	23/04/2025 14:51:09.069	false	192.168.40.147	192.168.118	40231	192.168.13	53	udp	IN	A	backup.local
4	23/04/2025 14:51:09.069	false	alias.fileserv.localhost	192.168.117	3446	192.168.12	53	udp	IN	CNAME	fileserv.localhost
5	23/04/2025 14:51:09.069	false	192.168.117.in-addr.arpa	192.168.117	25615	192.168.12	53	udp	IN	PTR	google.com
6	23/04/2025 14:51:09.069	false	192.168.117.in-addr.arpa	192.168.117	16271	192.168.12	53	udp	IN	PTR	google.com
7	23/04/2025 14:51:09.069	false	192.168.208.219	192.168.116	63770	192.168.12	53	udp	IN	PTR	google.com

Step 5 – Configure Index

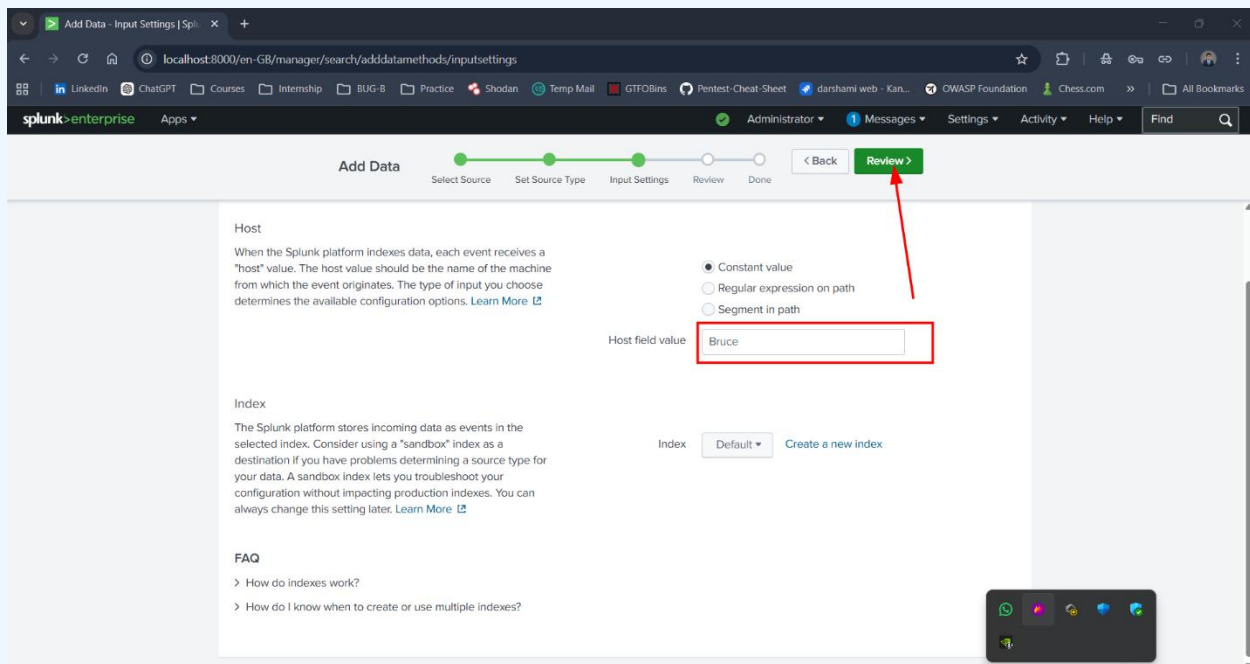
Choose or create an index.

Example:

`dns_lab`

Reason:

- Keeps lab data separated from other logs.
- Makes searching easier.



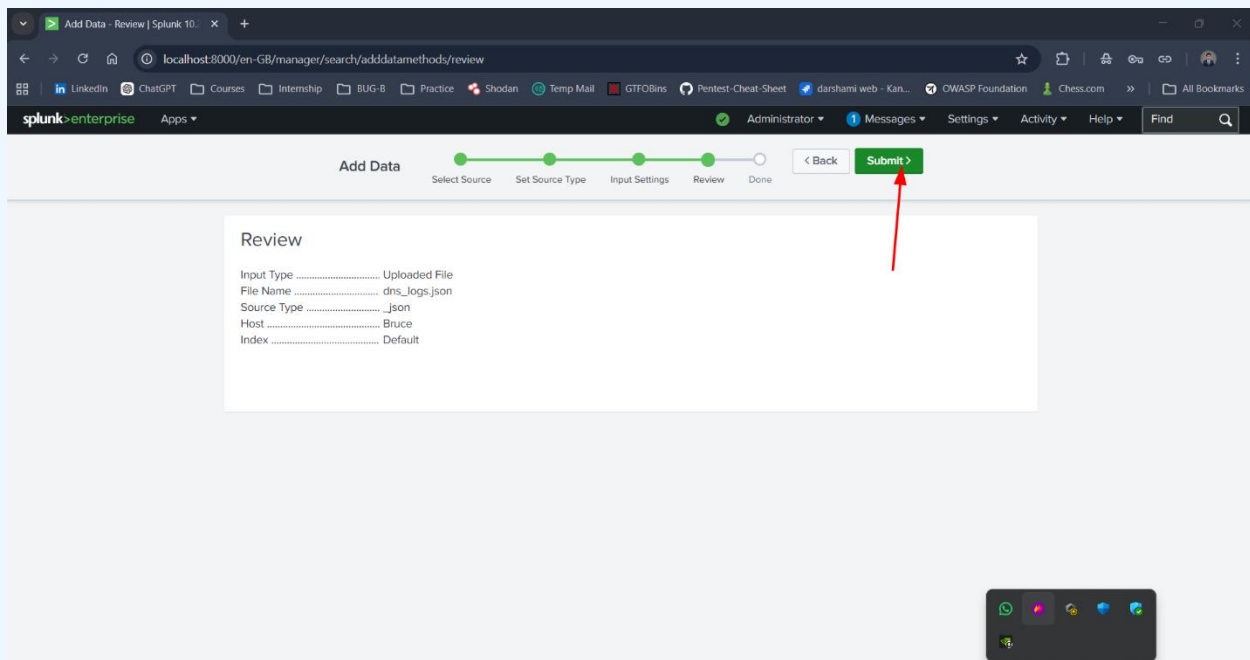
Step 6 – Review and Submit

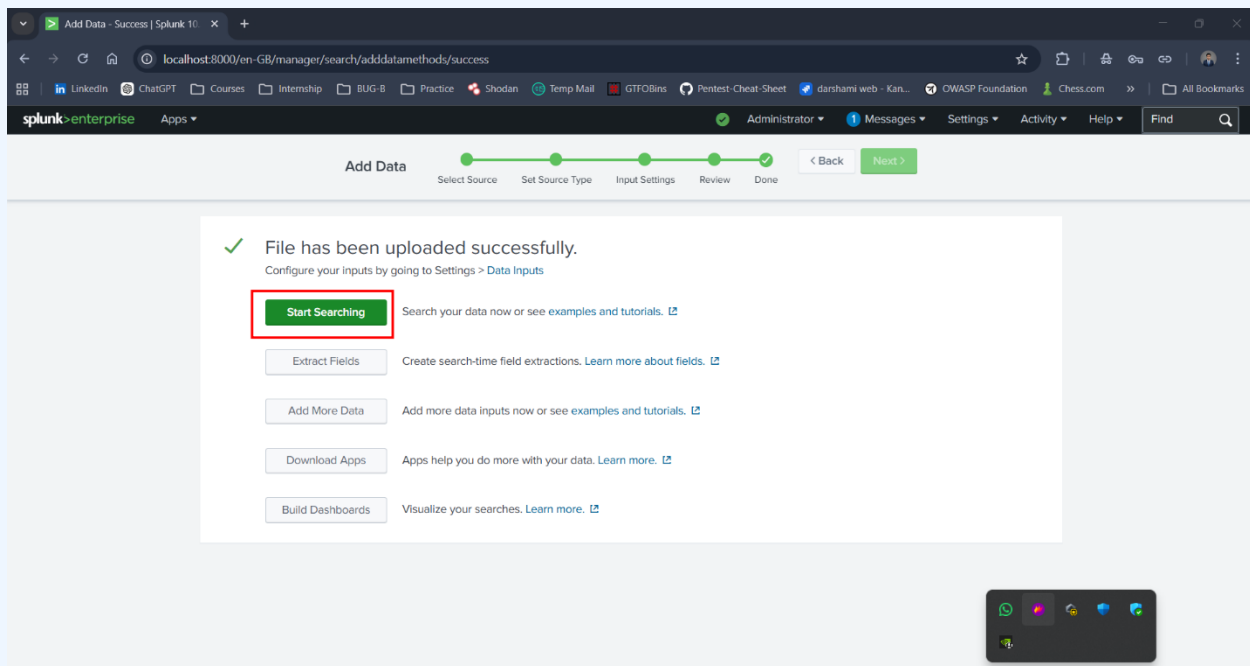
1. Review configuration.
2. Click **Submit**.
3. Splunk will index the data.

You can confirm ingestion by running:

```
index=dns_lab
```

If events appear, the upload was successful.





UNDERSTANDING DNS LOGS

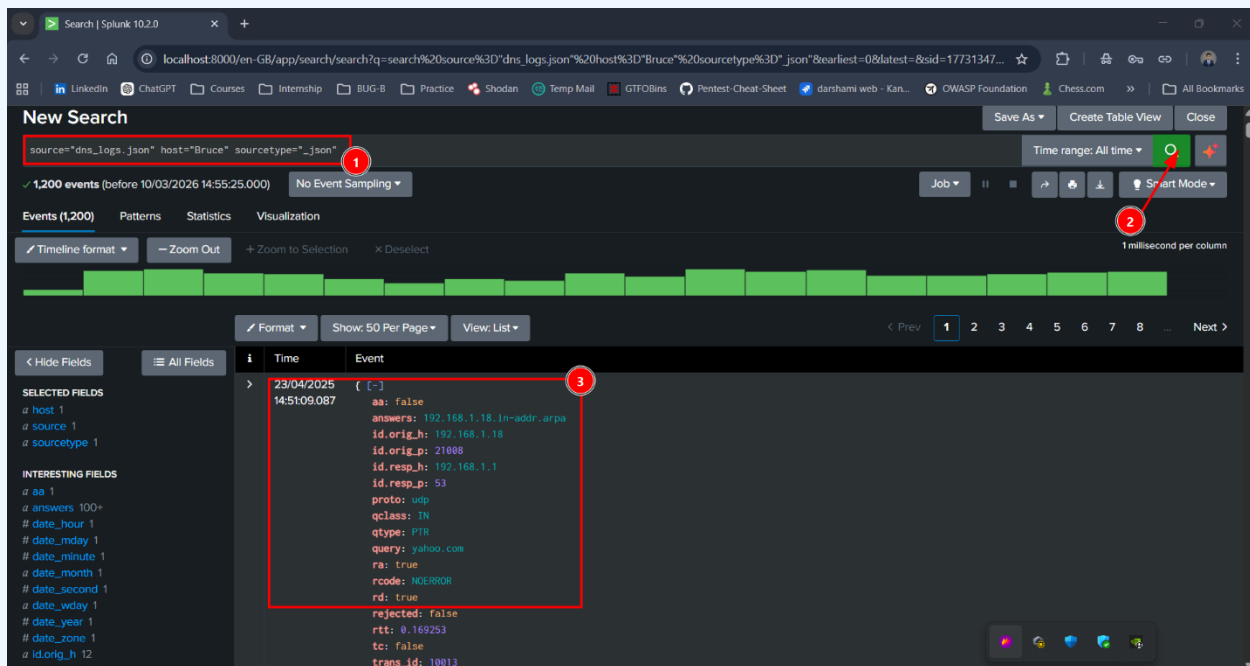
DNS logs record **domain name resolution requests** from clients.

Example DNS Event Fields:

Field	Description
id.orig_h	Source IP address making the DNS request
query	Domain name requested
qtype	Type of DNS query
answers	IP returned by DNS

Example Log:

```
{
  "id.orig_h": "192.168.1.10",
  "query": "google.com",
  "qtype": "A",
  "answers": "142.250.190.78"
}
```

LAB TASKS

Task 1: Identify the Most Frequently Queried Domain Names

SPL Query:

```
index=dns_lab sourcetype="json"
| stats count by query
| sort -count
```

Explanation

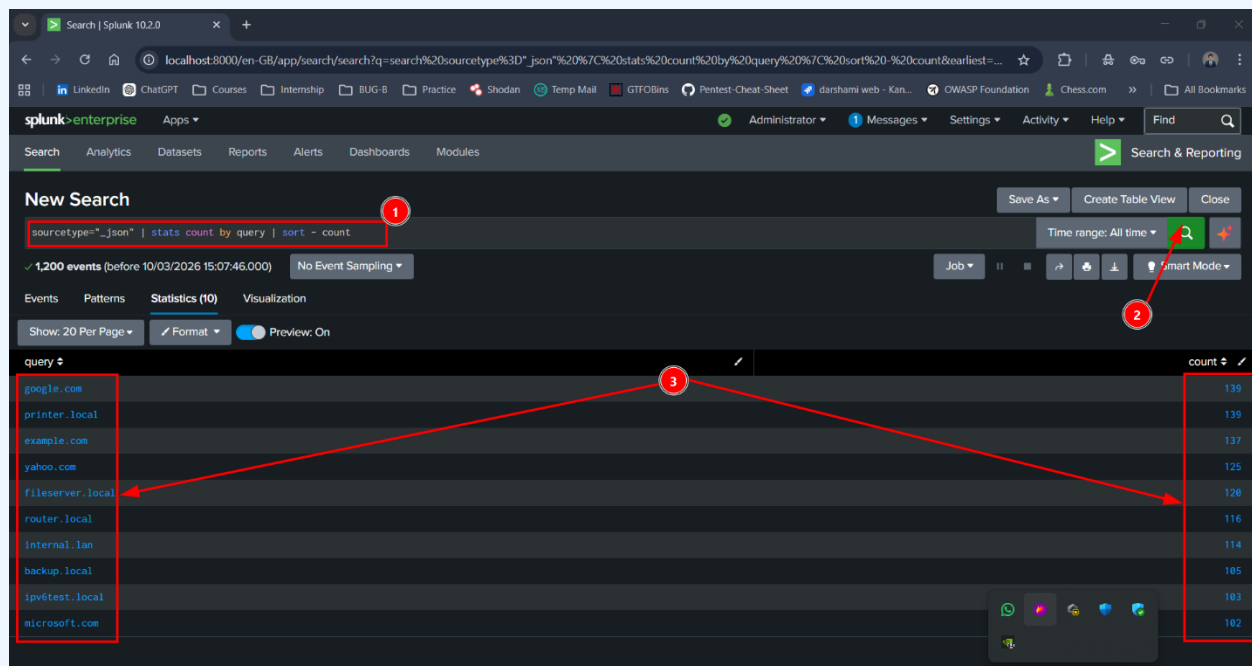
Command	Purpose
index=dns_lab	Searches logs stored in dns_lab index
sourcetype="json"	Filters only JSON formatted logs
stats count by query	Counts how many times each domain was queried
sort -count	Sorts results in descending order

What This Shows

This query reveals:

- Most visited websites
- Potential malware domains

- Unusual DNS traffic patterns



Task 2: Find the Most Active User IPs Generating DNS Traffic

SPL Query

```
index=dns_lab sourcetype="json"
| stats count by "id.orig_h"
| sort -count
```

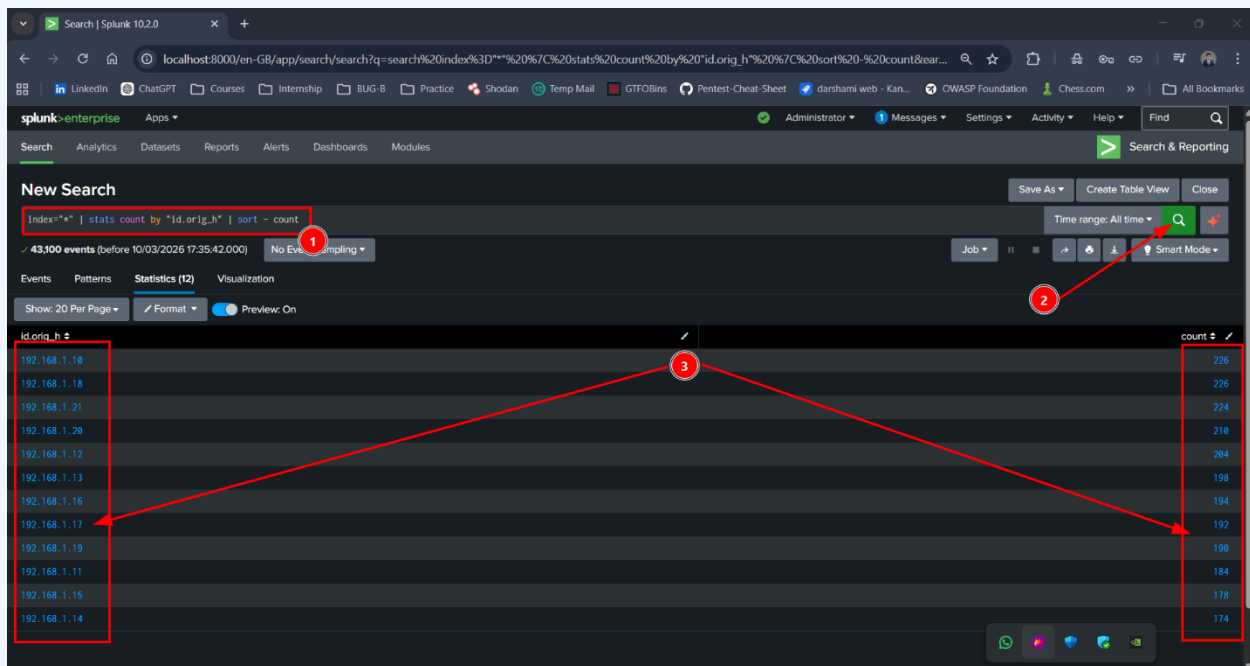
Explanation

Command	Purpose
stats count by "id.orig_h"	Counts DNS queries from each source IP
sort -count	Shows most active IPs first

What This Shows

This helps identify:

- Devices generating high DNS traffic
- Potential infected machines
- Misconfigured systems



If one host generates **extremely high DNS queries**, it may indicate **malware beaconing**.

Task 3: Breakdown of DNS Query Types

DNS queries can request different types of records.

SPL Query

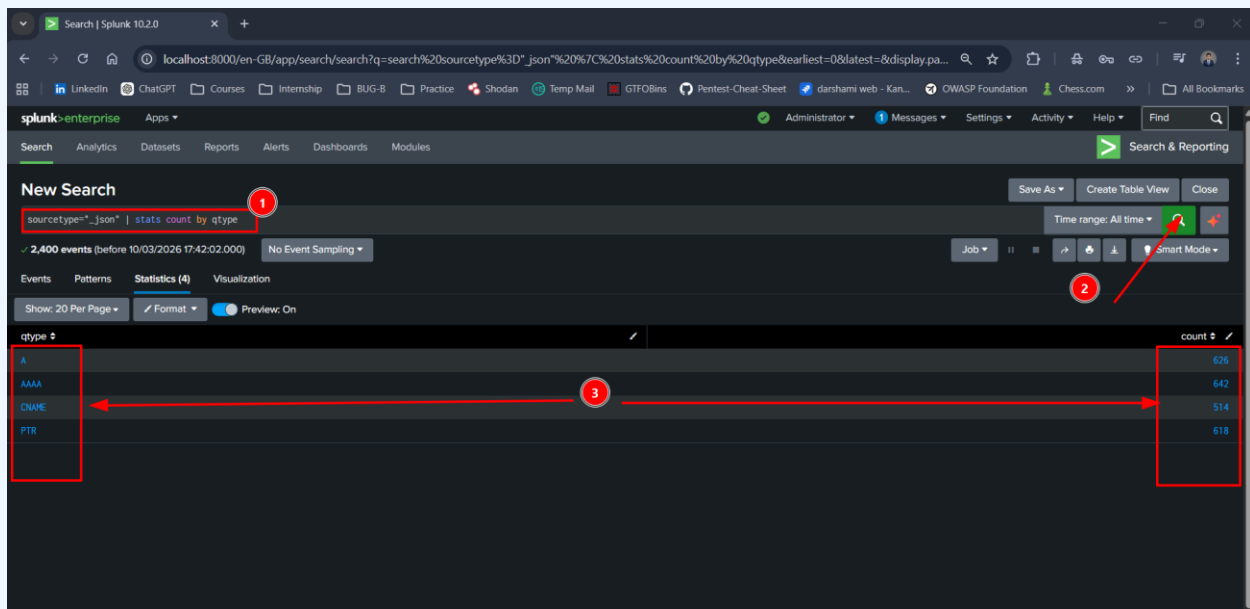
```
index=dns_lab sourcetype="json"
| stats count by qtype
```

Explanation

Command	Purpose
<code>stats count by qtype</code>	Groups DNS events by query type

Common DNS Query Types

Query Type	Meaning
A	Resolves domain to IPv4 address
AAAA	Resolves domain to IPv6 address
CNAME	Alias record
PTR	Reverse DNS lookup



SECURITY INSIGHT

DNS analysis helps detect:

Suspicious Activities

- Malware contacting command and control servers
- DNS tunneling
- Domain generation algorithms (DGA)
- Data exfiltration

Example Indicators:

- Very long domain names
- High number of DNS queries from one host
- Queries to unknown domains

Splunk makes it easier to **visualize and detect such anomalies**.

CONCLUSION

In this lab we successfully:

- Uploaded **Zeek DNS logs into Splunk**
- Learned how Splunk parses **JSON structured data**
- Used **SPL queries** to analyze DNS traffic

The analysis helped identify:

- Most frequently queried domains
- Most active source IPs
- Distribution of DNS query types

This type of analysis is essential for **network monitoring, threat detection, and incident response** in cybersecurity environments.